

The short answer is **no, they are not exactly the same.** [1]

While they use the same fundamental atmospheric profile inputs (temperature, water vapor mass, and viewing angle), SARTA mathematically splits **water vapor line absorption** and **water vapor continuum absorption** into two entirely separate, isolated calculation loops with different polynomial terms. [1]

The mathematical differences between the two sets highlight the distinct physics each one isolates:

## 1. Water Vapor Line Predictors (The 11-Term Set From Earlier)

This set handles **molecular quantum transitions** (where a single water molecule absorbs a very specific photon wavelength and vibrates or rotates).

- **The Math:** Heavily features the **square-root term** ( $\sqrt{W_r}$ ) to model "line saturation." When a specific spectral line absorbs as much light as it physically can, adding more water vapor barely blocks any more light. The square root linearizes this flattening curve.

## 2. Water Vapor Continuum Predictors (A Targeted 4-to-5 Term Set)

The continuum handles smooth, broad background absorption across wide swaths of the infrared spectrum. This is not caused by single isolated molecules, but rather by **molecular collisions** (water molecules bumping into each other, known as *self-continuum*, or water molecules bumping into nitrogen/oxygen, known as *foreign-continuum*).

Because it is a collision-based phenomenon, the physics change dramatically, and SARTA alters its predictor matrix to focus on: [2, 3, 4, 5, 6]

- **Pressure Products:** It incorporates localized layer pressure terms ( $P^2$ ) directly into the continuum predictors because higher pressure forces more frequent molecular collisions.
- **Squared Abundance terms:** Instead of scaling with a stabilizing square root ( $\sqrt{W_r}$ ), the self-continuum scales with the **square of the water vapor amount** ( $W_r^2$ ). If you double the number of water molecules in a layer, the number of collisions between them quadruples. SARTA's continuum loop specifically isolates these high-power terms to track this exponential drop in visibility. [3, 7]

## Why split them up?

By isolating the line predictors from the continuum predictors, SARTA can upgrade its background physics (for example, switching its underlying continuum baseline to modern standards like the [MT\\_CKD model](#)) without rewriting the entire 2,378-channel gas absorption framework. [8, 9]

Would you like to examine how SARTA combines these separate line and continuum values into a final **total water vapor transmittance**, or look into how **atmospheric pressure** scales these

collisions?

- [1] <https://www.researchgate.net>
- [2] <https://www.sciencedirect.com>
- [3] <https://convergence.cosmos.ru>
- [4] <https://www.nature.com>
- [5] <https://www.sciencedirect.com>
- [6] <https://agupubs.onlinelibrary.wiley.com>
- [7] <https://www.researchgate.net>
- [8] <https://www.gfdl.noaa.gov>
- [9] <https://journals.ametsoc.org>